

# Qatar-1b

R-1359

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-1\_210519 · created 2026-07-31T23:29:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2459353.8938 +/- 0.0015 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.158 +/- 0.018                 |
| Transit depth (Rp/Rs)^2                           | 2.5 +/- 0.58 [%]                |
| Orbital Inclination (inc)                         | 83.38 +/- 0.83                  |
| Airmass coefficient 1 (a1)                        | 0.992 +/- 0.014                 |
| Airmass coefficient 2 (a2)                        | 0.006 +/- 0.02                  |
| Scatter in the residuals of the lightcurve fit is | 1.46 %                          |
| Best Comparison Star                              | #3 - [379, 55]                  |
| Optimal Aperture                                  | 3.29                            |
| Optimal Annulus                                   | 10.82                           |
| Transit Duration (day)                            | 0.0656 +/- 0.0055               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.158 $\pm$ 0.018 vs 0.1463 (deviation 0.58 $\sigma$ ) |
| Duration fit vs published            | 0.0656 d vs 0.0692 d (deviation 0.58 $\sigma$ )        |
| Mid-time O-C                         | -4.4 min                                               |
| Depth SNR                            | 7.9                                                    |
| Residual scatter                     | 1.46 %                                                 |

## Light curve

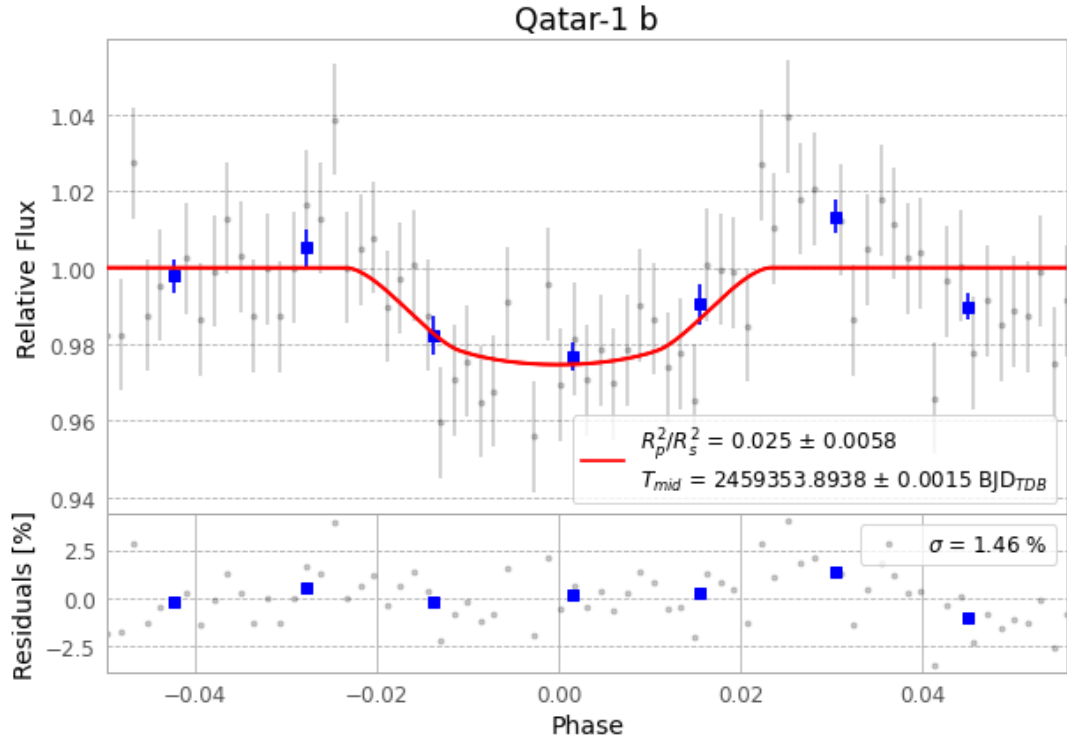


Figure 1 — FinalLightCurve\_Qatar-1 b\_2021-05-19.png (SHA-256 1b7f1c8164ab5537...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [339, 306], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 e06c592a40ca0c16...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

73 FITS frames (48 MB), Qatar-1210519074512.FITS → Qatar-1210519112112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-210519.log (9bc2ccc7994b...), FinalLightCurve\_Qatar-1 b\_2021-05-19.png (1b7f1c8164ab...), FinalLightCurve\_Qatar-1 b\_2021-05-19.csv (f89c5b8f110b...)

## Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 23:29Z.